

VortexTech Week 4 Advanced

Telecom Customer Churn — Retention Opportunities

Business question: Why are customers leaving, which groups are most at risk, and where should the business focus retention efforts?

Executive Summary

Customer churn is a major risk for subscription businesses because each customer who leaves represents lost recurring revenue. This analysis reviews 7,043 unique telecom customers. The overall churn rate is about 24.2%. The strongest pattern is tenure: customers in their first 12 months churn at about 52.5%, compared with only 6.1% among customers with 49–72 months of tenure. Contract type is another important signal: month-to-month customers churn at about 29.6%, compared with 18.0% for two-year customers. Fiber-optic customers show about 30.5% churn versus 21.2% for DSL customers. Electronic-check customers churn at about 27.5%, making payment behavior another useful risk signal. These results suggest that retention should be targeted rather than generic. The business should strengthen first-year onboarding, encourage suitable customers to move toward longer contracts, and investigate fiber service experience and payment friction. Because the dataset is synthetic and the analysis is descriptive, the patterns should be treated as hypotheses for further business validation rather than proof of causation.

Data Cleaning

The raw file contained 7,045 rows. Two duplicate customer records were removed, leaving 7,043 unique customers. 12 missing TotalCharges values were converted to numeric missing values and filled using MonthlyCharges × tenure. The final analysis contains no duplicate customer IDs and no missing TotalCharges values.

Key Findings

Finding	Business signal
Early tenure	0–12 months: 52.5% churn vs 6.1% at 49–72 months
Contract type	Month-to-month: 29.6% vs 18.0% for two-year
Internet service	Fiber: 30.5% vs 21.2% for DSL
Payment method	Electronic check: 27.5% churn

Actionable Recommendations

1. Build a first-year retention program with onboarding, proactive check-ins, service education, and early satisfaction surveys.
2. Encourage longer-term contracts using transparent, value-based offers rather than relying only on discounts.
3. Investigate fiber customer experience and electronic-check payment friction, then test targeted service or retention interventions.

Prepared as an internship capstone. The included dataset is synthetic and contains no real customer information.